INTRODUCTION TO ELEMENTARY PARTICLES

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Introduction

ELEMENTARY PARTICLE PHYSICS

Elementary particle physics addresses the question. "What is matter made of?" on the most fundamental level-which is to say. on the smallest scale of size. It's a remarkable fact that matter at the subatomic level consists of tiny chunks, with vast empty spaces in between. Even more remarkable, these tiny chunks come in a small number of different types (electrons, protons, neutrons, pi mesons, neutrinos, and so on). which are then replicated in astronomical quantities to make all the "stuff" around us. And these replicas are absolutely perfect copies-not just "pretty similar," like two Fords coming off the same assembly line. but utterly *indistinguishable.* You can't stamp an identification number on an electron, or paint a spot on it-if you've seen one, you've seen them all. This quality of absolute identicalness has no analog in the macroscopic world. (In quantum mechanics it is reflected in the Pauli exclusion principle.) It enormously simplifies the task of elementary particle physics: we don't have to worry about big electrons and little ones, or new electrons and old ones-an electron is an electron is an electron. It didn't have to be so easy.

My first job, then. is to introduce you to the various kinds of elementary particles. the actors, if you will. in the drama. I could simply *list* them, and tell you their properties (mass, electric charge, spin, etc.). but I think it is better in this case to adopt a historical perspective, and explain how each particle first came on the scene. This will serve to endow them with character and personality, making them easier to remember and more interesting to watch. Moreover, some of the stories are delightful in their own right.

Once the particles have been introduced. in Chapter 1. the issue becomes, "How do they interact with one another?" This question. directly or indirectly. will occupy us for the rest of the book. If you were dealing with two *macroscopic*

objects, and you wanted to know how they interact, you would probably begin by suspending them at various separation distances and measuring the force between them. That's how Coulomb determined the law of electrical repulsion between two charged pith balls. and how Cavendish measured the gravitational attraction of two lead weights. But you can't pick up a proton with tweezers or tie an electron onto the end of a piece of string; they're just too small. For practical reasons, therefore. we have to resort to less direct means to probe the interactions of elementary particles. As it turns out, almost all our experimental information comes from three sources: (1) scattering events, in which we fire one particle at another and record (for instance) the angle of deflection; (2) decays, in which a particle spontaneously disintegrates and we examine the debris; and (3) bound states, in which two or more particles stick together, and we study the properties of the composite object. Needless to say, determining the interaction law from such indirect evidence is not a trivial task. Ordinarily, the procedure is to guess a form for the interaction and compare the resulting theoretical calculations with the experimental data.

The formulation of such a guess ("model" is a more respectable term for it) is guided by certain general principles. in particular, special relativity and quantum mechanics. In the diagram below I have indicated the four realms of mechanics:

Small →

Fast ↓	
Classical mechanics	Quantum mechanics
Relativistic mechanics	Quantum field theory

The world of everyday life, of course, is governed by classical mechanics. But for objects that travel very fast (at speeds comparable to c), the classical rules are modified by special relativity, and for objects that are very small (comparable to the size of atoms, roughly speaking), classical mechanics is superseded by quantum mechanics. Finally, for things that are both fast *and* small, we require a theory that incorporates relativity and quantum principles: quantum field theory. Now, elementary particles *are* extremely small, of course, and typically they are also very fast. So elementary particle physics naturally falls under the dominion of quantum field theory.

Please observe the distinction here between a *type of mechanics* and a *particularforce law.* Newton's law of universal gravitation, for example, describes a specific interaction (gravity), whereas Newton's three laws of motion define a mechanical system (classical mechanics), which (within its jurisdiction) governs *all* interactions. The force law tells you what *F is,* in the case at hand; the mechanics tells you how to use *F* to determine the motion. The goal of elementary particle dynamics, then, is to guess a set of force laws which, within the context of quantum field theory, correctly describe particle behavior.

However, some general features of this behavior have nothing to do with the detailed form of the interactions. Instead they follow directly from relativity,

from quantum mechanics, or from the combination of the two. For example, in relativity, energy and momentum are always conserved, but (rest) mass is not. Thus the decay $\Delta \longrightarrow p + \pi$ is perfectly acceptable, even though the A weighs more than the sum of p plus π. Such a process would not be possible in classical mechanics, where mass is strictly conserved. Moreover, relativity allows for particles of zero (rest) mass-the very idea of a massless particle is nonsense in classical mechanics-and as we shall see, photons, neutrinos, and gluons are all (apparently) massless.

In quantum mechanics a physical system is described by its *state, s* (represented by the wave function ψ_s in Schrodinger's formulation, or by the *ket* $|s\rangle$ in Dirac's). A physical process, such as scattering or decay, consists of a *transition* from one state to another. But in quantum mechanics the outcome is not uniquely determined by the initial conditions: all we can hope to calculate. in general, is the *probability* for a given transition to occur. This indeterminacy is reflected in the observed behavior of particles. For example, the charged pi meson ordinarily disintegrates into a muon plus a neutrino, but occasionally one will decay into an *electron* plus a neutrino. There's no difference in the original pi mesons: they're all identical. It is simply a fact of nature that a given particle can go either way.

Finally, the union of relativity and quantum mechanics brings certain extra dividends that neither one by itself can offer: the existence of antiparticles, a proof of the Pauli exclusion principle (which in nonrelativistic quantum mechanics is simply an ad hoc hypothesis), and the so-called *TCP theorem.* I'll tell you more about these later on; my purpose in mentioning them here is to emphasize that these are features of the mechanical system itself, not of the particular model. Short of a catastrophic revolution, they are untouchable. By the way, quantum field theory in all its glory is difficult and deep, but don't be alarmed: Feynman invented a beautiful and intuitively satisfying formulation that is not hard to learn; we'11 come to that in Chapter 6. (The *derivation* of Feynman's rules from the underlying quantum field theory is a different matter, which can easily consume the better part of an advanced graduate course, but this need not concern us here.)

In the last few years a theory has emerged that describes all of the known elementary particle interactions except gravity. (As far as we can tell. gravity is much too weak to play any significant role in ordinary particle processes.) This theory–or, more accurately, this collection of related theories, incorporating quantum electrodynamics. the Glashow-Weinberg-Salam theory of electroweak processes, and quantum chromodynamics-has come to be called the *Standard Model.* No one pretends that the Standard Model is the final word on the subject, but at least we now have (for the first time) a full deck of cards to play with. Since 1978, when the Standard Model achieved the status of "orthodoxy," it has met every experimental test. It has, moreover, an attractive aesthetic feature: in the Standard Mode1 all of the fundamental interactions derive from a single genera1 principle, the requirement of *focal gauge invariance.* It seems likely that future developments will involve extensions of the Standard Model, not its repudiation. This book might be called an "Introduction to the Standard Model."

As that alternative title suggests. this is a book about elementary particle *theory,* with very little on experimental methods or instrumentation. These are important matters, and an argument can be made for integrating them into a text such as this. but they can also be distracting and interfere with the clarity and elegance of the theory itself. (I encourage you to read about experimental aspects of the subject. and from time to time I will refer you to particularly accessible accounts.) For now. I'll confine myself to scandalously brief answers to the two most obvious experimental questions.

HOW DO YOU *PRODUCE* ELEMENTARY PARTICLES?

Electrons and protons are no problem: these are the stable constituents of ordinary matter. To produce electrons one simply heats up a piece of metal. and they come boiling off. If one wants a beam of electrons. one then sets up a positively charged plate nearby, to attract them over, and cuts a small hole in it: the electrons that make it through the hole constitute the beam. Such an *electron gun* is the starting element in a television tube or an oscilloscope or an electron accelerator (Fig. 1.1).

To obtain protons you ionize hydrogen (in other words. strip off the electron). In fact, if you're using the protons as a *target.* you don't even need to bother about the electrons: they're so light that an energetic particle coming in will knock them out of the way. Thus, a tank of hydrogen is essentially a tank of protons. For more exotic particles there are three main sources: cosmic rays, nuclear reactors, and particle accelerators.

Cosmic Rays The earth is constantly bombarded with high-energy particles (principally protons) coming from outer space. What the source of these particles might be remains something of a mystery; at any rate. when they hit atoms in the upper atmosphere they produce showers of secondary particles (mostly muons, by the time they reach ground level), which rain down on us all the time. As a source of elementary particles. cosmic rays have two virtues: they are free. and their energies can be enormous – far greater than we could possibly produce in the laboratory. But they have two major disadvantages: The rate at which they strike any detector of reasonable size is very low. and they are completely uncontrollable. So cosmic ray experiments call for patience and luck.

Nuclear Reactors When a radioactive nucleus disintegrates. it may emit a variety of particles-neutrons. neutrinos. and what used to be called alpha rays (actually, alpha *particles,* which are bound states of two neutrons plus two protons), beta rays (actually, electrons or positrons), and gamma rays (actually, photons).

Particle Accelerators You start with electrons or protons, accelerate them to high energy, and smash them into a target. By skillful arrangements of absorbers and magnets, you can separate out of the resulting debris the particle species you wish to study. Nowadays it is possible in this way to generate intense sec-



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Figure 1.1 The Stanford Linear Accelerator Center (SLAC). Electrons and positrons are accelerated down a straight tube 2 miles long, reaching energies as high as 45 GeV. (Photo courtesy of SLAC.)

ondary beams of positrons, muons, pions. kaons. and antiprotons, which in turn can be fired at another target. The stable particles-electrons, protons, positrons, and antiprotons-can even by fed into giant *storage rings* in which, guided by powerful magnets, they circulate at high speed for hours at a time, to be extracted and used at the required moment (Fig. 1.2).

In general, the heavier the particle you want to produce, the higher must



Figure I.2 CERN, outside Geneva, Switzerland. SPS is the 450 GeV Super Proton Synchrotron, later modified to make a proton-antiproton collider; LEP is a 50 GeV electron-positron storage ring now under construction. (Photo courtesy of CERN.)

be the energy of the collision. That's why, historically, lightweight particles tend to be discovered first, and as time goes on, and accelerators become more powerful, heavier and heavier particles are found. At present, the heaviest known particle is the Z^0, with nearly 100 times the mass of the proton. It turns out that the particle gains enormously in energy if you collide two high-speed particles head-on, as opposed to firing one particle at a stationary target. (Of course. this calls for much better aim!) Therefore. most contemporary experiments involve colliding beams from intersecting storage rings; if the particles miss on the first pass, they can try again the next time around. Indeed, with electrons and positrons (or protons and antiprotons) the *same* ring can be used, with the plus charges circulating in one direction and the minus charges in the other.

There is another reason why particle physicists are always pushing for higher energies: In general, the higher the energy of the collision, the *closer* the two particles come to one another. So if you want to study the interaction at very short range, you need very energetic particles. In quantum-mechanical terms, a particle of momentum p has an associated wavelength λ given by the de Broglie formula $\lambda = h/p$, where h is Planck's constant. At large wavelengths (low momenta) you can only hope to resolve relatively large structures: in order to examine something extremely small, you need comparably short wavelengths. and hence high momenta. If you like. consider this a manifestation of the uncertainty principle ($\Delta x\, \Delta p \geq h/4\pi$)—to make Δx small, Δp must be large. However you

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look at it, the conclusion is the same: to probe *small distances* you need *high energies*.

HOW DO YOU *DETECT* ELEMENTARY PARTICLES?

There are many kinds of particle detectors-Geiger counters, cloud chambers, bubble chambers, spark chambers, photographic emulsions, Cerenkov counters, scintillators, photomultipliers, and so on (Fig. 1.3). Actually, a typical modem detector has whole arrays of these devices, wired up to a computer that tracks the particles and displays their trajectories on a television screen (Fig. 1.4). The details do not concern us, but there is one thing to be aware of: Most detection mechanisms rely on the fact that when high-energy charged particles pass through matter they ionize atoms along their path. The ions then act as "seeds" in the formation of droplets (cloud chamber) or bubbles (bubble chamber) or sparks (spark chamber), as the case may be. But electrically *neutral* particles do not cause ionization. and they leave no tracks. If you look at the bubble chamber photograph in Fig. 1.11, for instance, you will see that the five neutral particles are "invisible": their paths have been reconstructed by analyzing the tracks of the *charged* particles in the picture and invoking conservation of energy and momentum at each vertex. Notice also that most of the tracks in the picture are *curved* (actually, *all* of them are, to some extent; try holding a ruler up to one you think is straight). Evidently the bubble chamber was placed between the poles of a giant magnet. In a magnetic field B, a particle of charge q and momentum p will move in a circle of radius R given by the famous *cyclotron fromula:* $R = pc/qB$, where c is the speed of light. The curvature of the track in a known

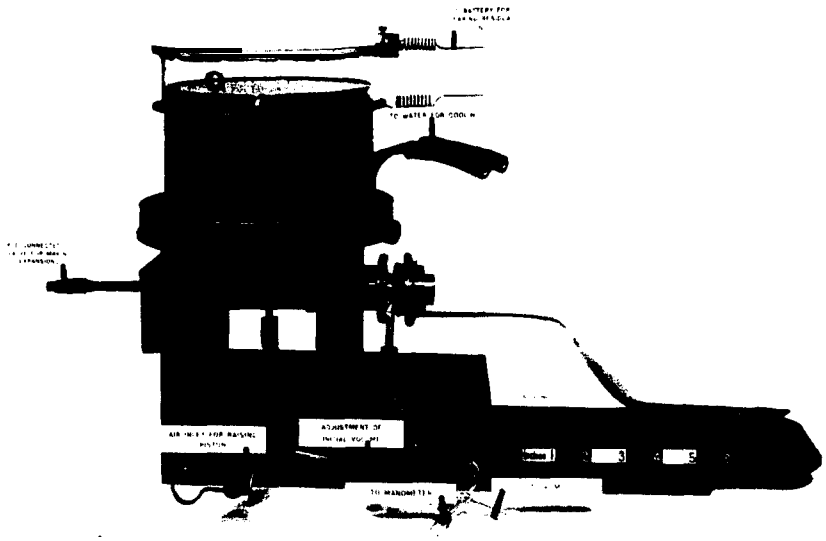


Figure I.3 An early particle detector: Wilson's cloud chamber (ca. 1900). (Photo courtesy Science Museum, London.)

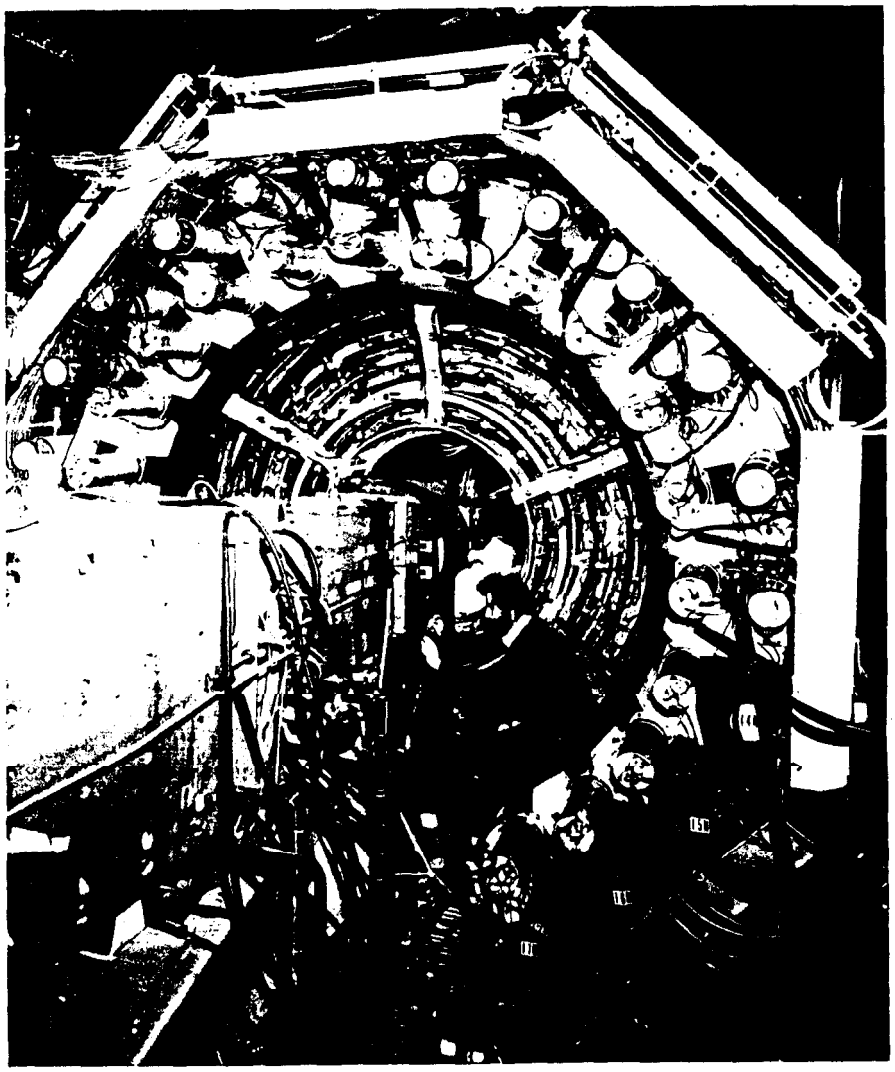


Figure 1.4 A modem particle detector: The Mark 1, at SLAC. (Photo courtesy SLAC.)

magnetic field thus affords a very simple measure of the particle's momentum. Moreover, we can immediately tell the sign of the charge from the *direction* of the curve.

UNITS

Elementary particles are small. so for our purposes the *normal* mechanical units – grams, ergs, joules. and so on – are inconveniently large. Atomic physicists introduced the *electron volt* – the energy acquired by an electron when accelerated



to courtesy SLAC.)

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through a potential difference of 1 volt: 1 eV = 1.6 X 10^{-19} joules. For us the eV is inconveniently *small,* but we're stuck with it. Nuclear physicists use keV (10^3 eV); typical energies in particle physics are MeV (10^6 eV), GeV (10^9 eV), or even TeV (10^{12} eV). Momenta are measured in MeV/c (or GeV/c, or whatever). and masses in MeV/c^2. Thus the proton weighs 938 MeV/c^2 = 1.67 X 10^{-24} g.

Actually, particle theorists are lazy (or clever, depending on your point of view)-they seldom include the c's and $\hbar$'s ($\hbar \equiv h/2\pi$) in their formulas. You're just supposed to fit them in for yourself at the end. to make the dimensions come out right. As they say in the business, "set $c = \hbar = 1$." This amounts to working in units such that time is measured in centimeters and mass and energy in inverse centimeters: the unit of time is the time it takes light to travel 1 centimeter, and the unit of energy is the energy of a photon whose wavelength is 2π centimeters. Only at the end of the problem do we revert to conventional units. This makes everything look very elegant, but I thought it would be wiser in this book to keep all the c's and $\hbar$'s where they belong, so that you can check for dimensional consistency as you go along. (If this offends you, remember that it is easier for you to ignore an A you don't like than for someone else to conjure one up in just the right place.)

Finally. there is the question of what units to use for electric charge. In introductory physics courses most instructors favor the *SI* system, in which charge is measured in *coulombs,* and Coulomb's law reads

$$F = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2} \qquad \text{(SI)}$$

Most advanced work is done in the *Gaussian* system, in which charge is measured in *electrostatic units* (esu), and Coulomb's law is written

$$F = \frac{q_1 q_2}{r^2} \qquad \text{(G)}$$

But elementary particle physicists prefer the *Heaviside-Lorentz* system, in which Coulomb's law takes the form

$$F = \frac{1}{4\pi}\frac{q_1 q_2}{r^2} \qquad \text{(HL)}$$

The three units of charge are related as shown:

$$q_{\mathrm{HL}} = \sqrt{4\pi}\, q_{\mathrm{G}} = \frac{1}{\sqrt{\epsilon_0}}\, q_{\mathrm{SI}}$$

In this book I shall use Gaussian units exclusively, in order to avoid unnecessary confusion in an already difficult subject. Whenever possible I will express results in *terms* of the *fine structure constant*

$$\alpha = \frac{e^2}{\hbar c} = \frac{1}{137}$$

where e is the charge of the electron in Gaussian units. Most elementary particle

texts write this as $e^2/4\pi$, because they are measuring charge in Heaviside-Lorentz units and setting $c = h = 1$; but everyone agrees that the *number* is $\frac{1}{137}$.

REFERENCES AND NOTES

This book is a brief survey of an enormous and rapidly changing subject. My aim is to introduce you to some important ideas and methods, to give you a sense of what's out there to be learned, and perhaps to stimulate your appetite for more. If you want to read further in quantum field theory, I particularly recommend:

Bjorken, J. D., and S. D. Drell. *Relativistic Quantum Mechanics* and *Relativistic Quantum Fields*. New York: McGraw-Hill, 1964.

Sakurai. J. J. *Advanced Quantum Mechanics*. Reading, MA: Addison-Wesley. 1967.

Itzykson, C., and J.-B. Zuber. *Quantum Field Theory*. New York: McGraw-Hill, 1980.

I warn you, however. that these are all difficult and advanced books. For elementary particle physics itself. the following books (listed in order of increasing difficulty) are especially useful:

Gottfried, K.. and V. F. Weisskopf. *Concepts of Particle Physics*. Oxford: Oxford University Press, 1984.

Frauenfelder, H., and E. M. Henley. *Subatomic Physics*. Englewood Cliffs. NJ: Prentice-Hall, 1974.

Perkins, D. H. *Introduction to High-Energy Physics*, 2d Ed. Reading, MA: Addison-Wesley, 1982.

Halzen, F., and A. D. Martin. *Quarks and Leptons*. New York: Wiley. 1984.

Aitchison, I. J. R., and A. J. G. Hey. *Gauge Theories in Particle Physics*. Bristol: Adam Hilger Ltd., 1982.

Close, F. E. *An Introduction to Quarks and Partons*. London: Academic. 1979.

Quigg, C. *Gauge Theories q/the Strong, Weak. and Electromagnetic Interactions*. Reading. MA: Benjamin/Cummings, 1983.

Cheng, T.-P.. and L.-F. Li. *Gauge Theories of Elementary Particle Physics*. New York: Oxford University Press, 1984.

$$d\Gamma = \left(\frac{g_W}{4\pi M_W c}\right)^4 \frac{m_\mu c}{\hbar} \frac{d^3\mathbf{p}_4}{E_4^2} \int_{1/2m_\mu c^2 - E_4}^{1/2m_\mu c^2} E_2(m_\mu c^2 - 2E_2)dE_2$$

$$= \left(\frac{g_W}{4\pi M_W c}\right)^4 \frac{m_\mu c}{\hbar} \left(\frac{m_\mu c^2}{2} - \frac{2}{3}E_4\right) d^3\mathbf{p}_4 \tag{10.34}$$

Finally, writing

$$d^3\mathbf{p}_4 = 4\pi\left(\frac{E_4}{c}\right)^2 \frac{dE_4}{c}$$

and dropping the subscript ($E \equiv E_4$ is the electron energy), we obtain

$$\frac{d\Gamma}{dE} = \left(\frac{g_W}{M_W c}\right)^4 \frac{m_\mu^2 E^2}{2\hbar(4\pi)^3}\left(1 - \frac{4E}{3m_\mu c^2}\right) \tag{10.35}$$

This tells us the energy distribution of the electrons emitted in muon decay. It tits the experimental spectrum perfectly (Fig. 10.1). The total decay rate is

$$\Gamma = \left(\frac{g_W}{M_W c}\right)^4 \frac{m_\mu^2}{2\hbar(4\pi)^3} \int_0^{1/2m_\mu c^2} E^2\left(1 - \frac{4E}{3m_\mu c^2}\right)dE$$

$$= \left(\frac{m_\mu g_W}{M_W}\right)^4 \frac{m_\mu c^2}{12\hbar(8\pi)^3} \tag{10.36}$$

and hence the lifetime of the muon is

$$\tau = \frac{1}{\Gamma} = \left(\frac{M_W}{m_\mu g_W}\right)^4 \frac{12\hbar(8\pi)^3}{m_\mu c^2} \tag{10.37}$$

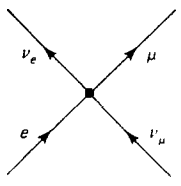
Notice that g_W and M_W do not appear *separately,* either in the muon lifetime formula or in the electron-neutrino scattering cross section; only their *ratio occurs.* It is traditional, in fact, to express weak interaction formulas in terms of the "Fermi coupling constant"

$$G_F \equiv \frac{\sqrt{2}}{8}\left(\frac{g_W}{M_W c^2}\right)^2 (\hbar c)^3 \tag{10.38}$$

Thus the muon lifetime is written

$$\tau = \frac{192\pi^3\hbar^7}{G^2 m_\mu^5 c^4} \tag{10.39}$$

In Fermi's original theory of beta decay (1933) there *was* no W; the interaction was supposed to be a direct four-particle coupling, represented in the Feynman language by a diagram of the form



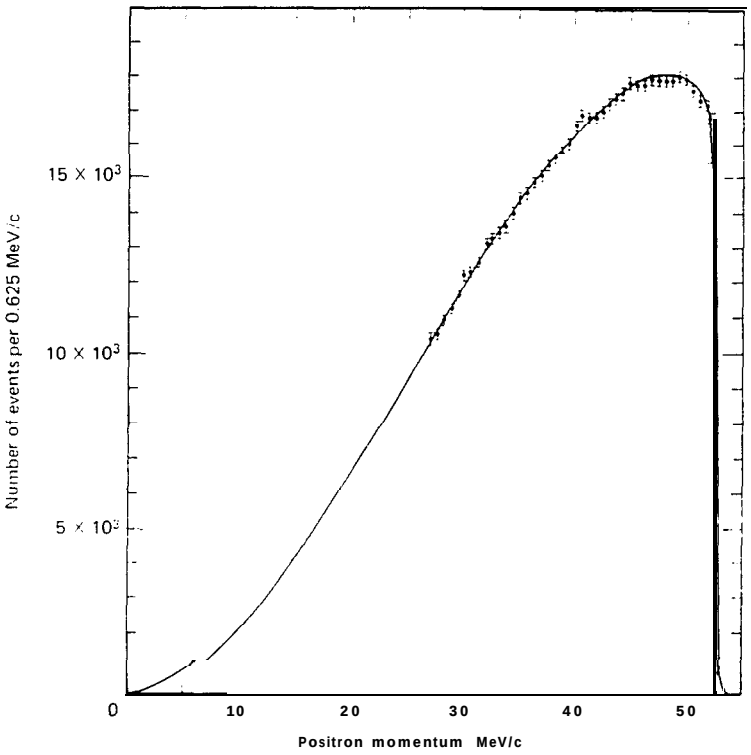
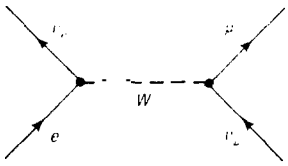


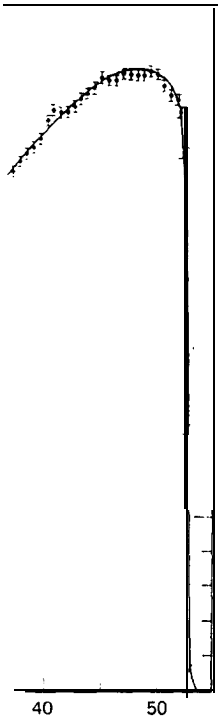
Figure 10.1 Experimental spectrum of positrons in $\mu^- \to e^+ + \nu_e + \nu_\mu$. The solid line is the theoretically predicted spectrum based on equation (10.35). corrected for electromagnetic effects. *(Source:* M. Bardon et al.. *Phys. Rev. Lett.* 14, 449 (1965).)

From the modern perspective. Fermi's theory combined the W propagator with the two vertex factors, in the diagram



to make an effective four-particle coupling constant G_F. It worked, but only because the W is so heavy that expression (10.4) is a good approximation to the true propagator (10.3),* and in fact it was recognized already in the fifties that Fermi's theory could not be valid at high energies. The idea of a weak mediator (analogous to the photon) was suggested by 0. Klein as far back as 1938.

* Fermi also thought the coupling was pure vector, as I mentioned in the footnote (*) on p. 303. Despite these defects (for which Fermi could scarcely be blamed: after all. he invented the theory at a time when the neutrino was a wild speculation and the Dirac equation itself was quite new) Fermi's theory was astonishingly prescient, and all subsequent developments have been relatively small adjustments to it.



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If we put in the observed muon lifetime and mass. we find that

$$G_F/(\hbar c)^3 = \frac{\sqrt{2}}{8}\left(\frac{g_w}{M_W c^2}\right)^2 = 1.166 \times 10^{-5}/\text{GeV}^2 \qquad (10.40)$$

The corresponding value of g_w (less accurately known. at present. because of the experimental uncertainty in M_W) is

$$g_w = 0.66 \qquad (10.41)$$

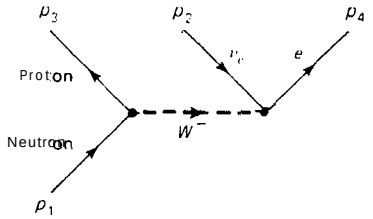
and hence the "weak fine structure constant" is

$$\alpha_w = \frac{g_w^2}{4\pi} = \frac{1}{29} \qquad (10.42)$$

This number should come as something of a shock: It is *larger* than the electromagnetic fine structure constant ($\alpha = \frac{1}{137}$), by a factor of nearly five! Weak interactions are feeble nor because the *intrinsic* coupling is small (it *isn't*), but because the *mediators* are so *massive*-or, more precisely. because we typically work at energies so far below the W mass that the denominator in the propagator $(q^2 - M_W^2 c^2)$ is extremely large. New machines are presently under construction that will run at energies close to $M_W c^2$, and in this regime the "weak" interactions will far surpass the electromagnetic ones in strength.

10.3 DECAY OF THE NEUTRON

The success of the muon decay formula (10.35) encourages us to apply the same methods to the decay of the neutron. $n \longrightarrow p + e + \bar{\nu}_e$. Of course. the neutron and proton are composite particles, but just as the Mott and Rutherford cross sections (which treat the proton as an elementary "Dirac" particle) give a good account of low-energy electron-proton scattering. so we might hope that the diagram



(the same as for muon decay, only with $n \longrightarrow p + W^-$ in place of $\mu \longrightarrow \nu_\mu + W^-$) will afford a reasonable approximation to neutron beta decay. From a calculational point of view the only new feature is that 3 is now a massive particle (a proton, instead of a neutrino). As it happens (Problem 10.4) this does not change the amplitude:

$$\langle |\mathcal{M}|^2 \rangle = 2\left(\frac{g_w}{M_W c}\right)^4 (p_1 \cdot p_2)(p_3 \cdot p_4) \qquad (10.43)$$



which is the same as (10.16). In the rest frame of the neutron. we find

$$\langle|\mathcal{M}|^2\rangle = \left(\frac{g_w}{M_W c}\right)^4 m_n E_2[(m_n^2 - m_p^2 - m_e^2)c^2 - 2m_n E_2] \tag{10.44}$$

In this case the electron rest energy is a substantial fraction of the total energy released, $(m_n - m_p - m_e)c^2$, so we *cannot* afford to ignore the electron mass.

The decay rate calculation proceeds as before:

$$d\Gamma = \frac{\langle|\mathcal{M}|^2\rangle}{2\hbar m_n}\left(\frac{c\,d^3\mathbf{p}_2}{(2\pi)^3 2E_2}\right)\left(\frac{c\,d^3\mathbf{p}_3}{(2\pi)^3 2E_3}\right)\left(\frac{c\,d^3\mathbf{p}_4}{(2\pi)^3 2E_4}\right)(2\pi)^4\,\delta^4(p_1 - p_2 - p_3 - p_4) \tag{10.45}$$

where $E_2 = c|\mathbf{p}_2|, \qquad E_3 = c\sqrt{\mathbf{p}_3^2 + m_p^2 c^2}, \qquad E_4 = c\sqrt{\mathbf{p}_4^2 + m_e^2 c^2}$ (10.46)

The $\mathbf{p}_3$ integral yields

$$d\Gamma = \frac{2c^3\langle|\mathcal{M}|^2\rangle}{(4\pi)^5\hbar m_n}\frac{d^3\mathbf{p}_2\,d^3\mathbf{p}_4}{E_2E_3E_4}\,\delta\left(m_n c - \frac{E_2}{c} - \frac{E_3}{c} - \frac{E_4}{c}\right) \tag{10.47}$$

which is the same as equation (10.23), except that this time

$$E_3 = c\sqrt{(\mathbf{p}_2 + \mathbf{p}_4)^2 + m_p^2 c^2} \tag{10.48}$$

To carry out the $\mathbf{p}_2$ integral. we set

$$d^3\mathbf{p}_2 = |\mathbf{p}_2|^2\,d|\mathbf{p}_2|\sin\theta\,d\theta\,d\phi = \frac{1}{c^3}E_2^2\,dE_2\sin\theta\,d\theta\,d\phi \tag{10.49}$$

and orient the coordinates so that the z axis lies along $\mathbf{p}_4$ (which is fixed, for purposes of the $\mathbf{p}_2$ integral); then

$$E_3 = c\sqrt{|\mathbf{p}_2|^2 + |\mathbf{p}_4|^2 + 2|\mathbf{p}_2||\mathbf{p}_4|\cos\theta + m_p^2 c^2} \equiv cx \tag{10.50}$$

and

$$\frac{E_2\sin\theta\,d\theta}{E_3} = -\frac{dx}{|\mathbf{p}_4|} \tag{10.51}$$

The ϕ and θ (or rather. x) integrals yield

$$d\Gamma = \frac{\langle|\mathcal{M}|^2\rangle}{(4\pi)^4\hbar m_n}\frac{dE_2\,d^3\mathbf{p}_4}{E_4|\mathbf{p}_4|}\,I \tag{10.52}$$

where

$$I \equiv \int_{x_-}^{x_+}\delta\left(m_n c - x - \frac{E_2}{c} - \frac{E_4}{c}\right)dx$$

$$= \begin{cases} 1, & \text{if } x_- < \left(m_n c - \frac{E_2}{c} - \frac{E_4}{c}\right) < x_+ \\ 0, & \text{otherwise} \end{cases} \tag{10.53}$$

and the limits [from (10.50)] are

$$x_\pm = \sqrt{(|\mathbf{p}_2| \pm |\mathbf{p}_4|)^2 + m_p^2 c^2} \tag{10.54}$$

As before, equation (10.53) defines the range of the E_2 integral: I'll let you work out the algebra (Problem 10.5):

$$E_{\pm} = \frac{\frac{1}{2}(m_n^2 - m_p^2 + m_e^2)c^2 - m_n E_4}{m_n - E_4/c^2 \mp |\mathbf{p}_4|/c} \tag{10.55}$$

The E_2 integral is thus

$$\int_{E_-}^{E_+} E_2[(m_n^2 - m_p^2 - m_e^2)c^2 - 2m_n E_2]dE_2 \equiv J(E_4) \tag{10.56}$$

and since

$$d^3\mathbf{p}_4 = 4\pi|\mathbf{p}_4|^2\, d|\mathbf{p}_4| = \frac{4\pi}{c^2}\,|\mathbf{p}_4|E_4\, dE_4 \tag{10.57}$$

we conclude that

$$\frac{d\Gamma}{dE} = \frac{1}{\hbar c^2(4\pi)^3}\left(\frac{g_w}{M_w c}\right)^4 J(E) \tag{10.58}$$

(since there is no further occasion for ambiguity, I'm eliminating the subscript on E_4; from now on E is the electron energy).

Equation (10.58) is *exact* (use it. if you like, to rederive equation (10.35). setting $m_n \longrightarrow m_\mu$ and m_p. $m_e \longrightarrow 0$). but $J(E)$ is a rather cumbersome function. From the definition (10.56):

$$J(E) = \frac{1}{2}(m_n^2 - m_p^2 - m_e^2)c^2(E_+^2 - E_-^2) - \frac{2m_n}{3}(E_+^3 - E_-^3) \tag{10.59}$$

where $E_\pm$ are given by equation (10.55). It pays to approximate. at this stage, recognizing that there are four small numbers here:

$$\epsilon \equiv \frac{m_n - m_p}{m_n} = 0.0014, \qquad \delta \equiv \frac{m_e}{m_n} = 0.0005,$$

$$\eta \equiv \frac{E_4}{m_n c^2} \quad (\delta < \eta < \epsilon), \qquad \phi \equiv \frac{|\mathbf{p}_4|}{m_n c} \quad (0 < \phi < \eta) \tag{10.60}$$

(The last of these is not independent. of course: $\phi^2 = \eta^2 - \delta^2$.) Expanding to lowest order (Problem 10.5). we obtain

$$J \cong 4m_n^4 c^6 \eta\phi(\epsilon - \eta)^2 = \frac{A}{c^2} E\sqrt{E^2 - m_e^2c^4}[(m_n - m_n)c^2 - El^{\prime\prime} \tag{10.61}$$

So the distribution of electron energies is given by

$$\frac{d\Gamma}{dE} = \frac{1}{\pi^3\hbar}\left(\frac{g_w}{2M_w c^2}\right)^4 E\sqrt{E^2 - m_e^2c^4}[(m_n - m_p)c^2 - E]^2 \tag{10.62}$$

The experimental results are shown in Figure 10.2. The electron energies range from $m_e c^2$ up to about $(m_n - m_p)c^2$ (Problem 10.6). Integrating over E, we get the total decay rate (Problem 10.7):

$$\Gamma = \frac{1}{4\pi^3\hbar}\left(\frac{g_w}{2M_w c^2}\right)^4 (m_e c^2)^5$$

$$\times\left[\frac{1}{15}(2a^4 - 9a^2 - 8)\sqrt{a^2 - 1} + a\,\ln(a + \sqrt{a^2 - 1})\right] \tag{10.63}$$